PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

HostelSpace

Hostel Management System

SYNOPSIS / PROJECT PROPOSAL
(Deliverable 1)

Item	Details
Project Title	HostelSpace
Assigned Project	Hostel Management System
Group Number	Team 11
Course	UE24CS341A : Software Engineering
Development Methodology	Agile
Version	1.0
Date	1 September 2026

Team Profile

Sl. No.	USN	SRN / PRN	Student Name
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1. Proposed Project Description

HostelSpace is a simple web-based hostel management system for students and hostel staff. At present, many hostel activities are handled through notice boards, paper forms, registers, messages, and repeated visits to the hostel office. Information is spread across different places, so students may need to ask several people before they receive an update. Staff also have to check different records to answer common questions.

HostelSpace brings these activities into one portal. A student can sign in, view hostel notices and rules, check whether the study room is open, see the assigned laundry day, track clothes given and collected, apply for leave, submit attendance, raise a maintenance complaint, view mess details, pay hostel fees, and check deviation-weightage records. Students will see only their own records and the services available to them.

Hostel staff will use the same system according to their roles. Wardens will process leave, attendance, and discipline records. The Chief Warden will supervise the wardens and handle important cases. The Hostel Director will have overall control of hostel information and authorized users. Laundry, mess, and maintenance staff will update only the records related to their work. The project keeps every feature direct and limited so that the team can build, test, and explain the complete system within the course schedule.

2. Problem Statement

The main problem is that hostel information and services are not available in one place. A notice may be displayed on a board, a leave request may be written on paper, attendance may be stored in a separate register, and a complaint may be followed up through messages or direct visits. Students can miss notices or may not know whom to contact. They may also have to repeatedly ask whether a request was accepted, whether a complaint was resolved, whether a fee was recorded, or when their laundry was returned.

This situation also creates extra work for hostel staff. Wardens and service staff must search through registers or messages, the same information may be written more than once, and an old record can be difficult to find. When several wardens are involved, it may also be unclear which request is assigned to whom or which cases need the attention of the Chief Warden. The Hostel Director does not get one clear view of the important hostel records.

HostelSpace solves this problem by keeping the selected activities in one role-based system. Each student can view personal updates without seeing another student’s private records. Each staff member can update only the assigned work. Requests and service records will have a clear status, making follow-up easier for students, wardens, the Chief Warden, the Hostel Director, and other hostel staff.

3. Target Users and Stakeholders

User / Stakeholder	Role in the System
Student	Views hostel information, uses student services, submits requests, pays hostel fees, and tracks personal records.
Wardens	Multiple wardens review assigned leave requests, monitor attendance, and support discipline-related activities.
Chief Warden	Acts as the main warden, supervises the warden workflow, handles escalated matters, and reviews important attendance or discipline records.
Hostel Director	Has overall administrative oversight and manages authorized users, hostel information, rules, service records, and reports.
Maintenance Staff	Views assigned maintenance complaints and updates their progress and completion status.
Mess Staff	Maintains student mess allocation and digital coupon records.
Laundry Staff	Updates the tracker for allocated laundry days, clothes received, and clothes returned.

4. Objectives

• Provide a single portal for routine hostel information and student services.

• Reduce dependence on paper forms, registers, and repeated manual follow-up.

• Make notices, hostel contacts, and rules easy for students to access.

• Allow students to see whether the study room is currently open or closed.

• Track the laundry day assigned to each student and the handover and collection status.

• Provide clear workflows for leave requests, attendance, maintenance complaints, fee payments, mess records, and deviation weightage.

• Give wardens, the Chief Warden, the Hostel Director, and service staff access only to the functions relevant to their roles.

• Develop the system in small Agile iterations with clear team ownership.

5. Project Scope

HostelSpace will cover the essential hostel activities required for the first working version. The scope is kept simple so that every selected function can be completed, tested, and demonstrated by the team.

Student access: Students will sign in, view and update permitted profile details, read hostel notices and rules, and view their own service and discipline records.

Study-room status: Authorized staff will mark the study room as open or closed and may add the applicable timings or a short status message. The system will only display availability; it will not provide booking or seat reservation.

Laundry tracker: Laundry days will be allocated by hostel staff. Students will not select a day or book a slot. The module will show the assigned day and track clothes given, garment count or item details, clothes collected, and relevant timestamps.

Leave and attendance: Students will submit leave applications for approval by a warden. Attendance will use location verification only when the student actively submits attendance. Location will not be tracked continuously.

Other student services: Students will be able to submit maintenance complaints, view mess allocation and coupons, make or record hostel fee payments, and view deviation-weightage entries linked to hostel rules.

Administration: Wardens, the Chief Warden, the Hostel Director, and service staff will update only the records and workflows assigned to their roles. Reports will be limited to the information required for the project demonstration.

6. Functional Features

Feature	Planned Functionality
Student profile	Students can view their hostel profile and update only permitted contact or personal details.
Notices and rules	Authorized staff can publish notices and maintain hostel information, contacts, and rules. Students can view active information.
Study-room status	Students can see an open/closed indicator, timings, and an optional staff message. No booking function is included.
Laundry tracker	Staff assign the laundry day. The tracker records clothes received, item or garment count, return status, and handover/collection times.
Leave management	Students submit a reason and leave period. A warden approves or rejects the request, and the student sees the current decision.
Attendance	Students submit attendance during the allowed period. The system checks location at that moment and records the result.
Maintenance complaints	Students submit a category, description, and room details. Staff update the complaint from submitted to in progress and resolved.
Mess services	Students view their mess allocation and available digital coupon records; authorized staff maintain these records.
Fee payment	Students submit a hostel fee payment using the selected payment method and view payment status and receipt/reference details.
Deviation weightage	Authorized wardens record a rule violation and its weightage. The system totals the recorded weightage, and students view their own entries.

7. Functional Ownership and Team Responsibilities

Each member owns complete user-facing functionality, including the related screens, data handling, validation, testing, and documentation. All members will also participate in requirements, design discussions, Git commits, reviews, integration, and presentation.

Member	Primary Functional Ownership	Supporting Responsibility
Anvi Srinivas	Student profile; hostel notices; hostel information and contacts; hostel rules	Requirements review and consistency of student-facing screens
Arin Singh	Leave applications and warden approval; location-verified attendance; deviation-weightage records and rule references	Security checks and documentation of role-based workflows
Aryan Singh	Study-room open/closed status; allocated laundry-day tracker; clothes handover and collection records	Database field definitions and module-level testing
Atiksh Gour	Maintenance complaints; mess allocation and coupons; hostel fee payment and payment-status records	GitHub/Jira coordination and integration testing

8. Proposed Technology and Tools

The project will use a small, familiar web-development stack. Additional libraries will be introduced only when required for a selected function.

Purpose	Selected Technology / Tool
User interface and application	Next.js with JavaScript, HTML, and CSS
Server-side functions	Next.js server functions or simple API routes
Database	SQL database for project records
Authentication	Simple login verification with role-based access checks
Attendance location check	Browser location permission only during attendance submission
Version control	Git and GitHub
Agile planning	Jira
Testing	Basic automated tests and manual functional testing
Development environment	Standard code editor and web browser

9. Plan of Work

Sprint	Planned Output	Acceptance Evidence
Sprint 1	Project setup, Jira backlog, GitHub repository, simple authentication, user roles, and student profile	User stories, repository activity, and working sign-in/profile
Sprint 2	Notices, hostel information and rules, study-room open/closed status, and laundry-day allocation display	Working pages, sample records, tests, and pull requests
Sprint 3	Laundry handover/collection tracker, leave application, and warden approval workflow	Feature demonstration and functional test results
Sprint 4	Location-verified attendance, deviation-weightage records, and complaint submission/status workflow	Integration results and role-access checks
Sprint 5	Mess allocation/coupons, fee payment and payment-status records, and required reports	Completed workflows, validation evidence, and test summary
Sprint 6	Final integration, defect correction, security checks, documentation, and complete system demonstration	Stable build, final report evidence, and team demo

10. Initial Constraints and Assumptions

• HostelSpace will be accessed through a modern web browser.

• Only authorized users will receive staff roles; students will see only their own records and permitted services.

• There may be several wardens, while one Chief Warden acts as the main supervising warden.

• The Hostel Director will have overall administrative oversight of the system.

• The study-room module provides only open/closed information and does not support booking.

• Laundry days are assigned by staff; students cannot choose or reserve a laundry slot.

• Location permission is requested only when attendance is submitted and is not used for background tracking.

• The fee-payment module will store the amount, date, status, and transaction or receipt reference required for the project workflow.

• The system will use sample or authorized project data during development and demonstration.

• Scope changes after faculty feedback will be accepted only if every member retains complete functional ownership.